Cong Yu

ML Systems | Agentic Reinforcement Learning | Data Systems | Quantum Software
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Profile

Doctoral researcher at Aalto University building **ML and data systems** for LLM post-training, agentic reinforcement learning, latency-bounded analytics, and quantum software. First author at **PVLDB 2026** and **ACM CAIS 2026**, with prior engineering experience at TikTok, StarFive, and Baidu Quantum. Expected PhD completion in December 2027.

Technical Skills

ML and LLM Systems: PyTorch, Hugging Face Transformers, vLLM, veRL / VerlTool, Ray, FSDP, GRPO, SFT, agentic RL

GPU and Distributed Systems: multi-GPU training and inference, CUDA, CUDA-Q, cuQuantum, Slurm, Linux

Data Systems: C/C++, Python, complex event processing, SQL MATCH_RECOGNIZE, GraphRAG, graph databases, openCypher

Quantum Software: Qiskit, PennyLane, OpenQASM 3.0, circuit generation, simulation, optimization, and verification

Languages and Tools: Python, C/C++, Scala, OCaml, Git, Bash, CMake, Weights & Biases

Experience

Aalto University, Doctoral Researcher in Computer Science — Espoo, Finland — Dec 2023 - Present

- Lead systems research spanning LLM post-training, data management, and quantum software; first author at PVLDB 2026 and ACM CAIS 2026.
- Built multi-GPU agentic-RL and verifier pipelines around PyTorch, vLLM, Ray, FSDP, and Slurm, including paper-scale training of a 4B model on 8 GPUs.
- Developed open-source C++/Python research systems and reproducible artifacts, with public models and datasets released through GitHub and Hugging Face.
- Advisor: Prof. Francesco Croce. Expected PhD completion: December 2027.

TikTok Pte. Ltd., Backend Engineer Intern, Data Engine Development — Singapore — May 2023 - Sept 2023

StarFive Technology Co., Ltd., Engineer Intern — Singapore — Dec 2022 - Feb 2023

Institute for Quantum Computing, Baidu, Software Engineer Intern — Beijing, China — July 2020 - Sept 2020

University of Southern California, Directed Research Student — Los Angeles, CA — May 2019 - Sept 2019

University of Washington, Visiting Research Student (VISIT) — Seattle, WA — Sept 2017 - Nov 2017

Education

PhD **Aalto University**, Computer Science — Espoo, Finland — Dec 2023 - Dec 2027
- Advisor: Prof. Francesco Croce. Expected completion: December 2027.

National University of Singapore, Master of Computing (General Track) — Singapore — Jan 2022 - Nov 2023
- GPA: 4.08/5.00

MS **University of Southern California**, Electrical Engineering — Los Angeles, CA — Aug 2018 - Dec 2019
- GPA: 3.96/4.00; ECE Master's Students Honors Program

BE **Beihang University**, Electronic and Information Engineering — Beijing, China — Sept 2014 - July 2018
- GPA: 3.60/4.00

Selected Research Systems

Vista - Verifier-in-the-Loop Agentic RL May 2026

First author; ACM Conference on AI and Agentic Systems (CAIS 2026)

- Designed hierarchical verified rewards and budget-aware staged evaluation for OpenQASM synthesis; improved Pass@10 executability by 1.13x and semantic quality by 1.10x while reducing verification cost by 1.77x.
- Delivered a reproducible training, inference, and evaluation stack with a public 4B checkpoint and dataset.

SHARP - Shared-State Reduction for Pattern Matching May 2026

Co-first author; Proceedings of the VLDB Endowment, 19(5)

- Built a C++/Python state-management library with constant-time cost-based pruning across CEP, SQL row-pattern matching, and GraphRAG; retained 95%, 93%, and 72% recall under a 50% latency budget.
- Packaged three reproducible workload artifacts and validated the system on Ubuntu, SUSE Linux Enterprise, and Red Hat Enterprise Linux.

QUASAR - Tool-Augmented LLMs via Agentic RL Oct 2025

First author; arXiv preprint

- Combined quantum simulators with hierarchical RL rewards to post-train a 4B LLM for quantum assembly generation, reaching 99.31% Pass@1 and 100% Pass@10 validity.
- Outperformed GPT-4o, GPT-5, DeepSeek-V3, and supervised- or RL-only baselines on generated-circuit validity.

Agent-Q - LLM Fine-Tuning for Quantum Circuits Sept 2025

Co-author; IEEE Quantum Week (QCE 2025)

- Contributed to an end-to-end SFT pipeline and a 14,000-circuit dataset spanning 12 optimization problems and QAOA, VQE, and adaptive VQE circuits.

ZX-DB - Database-Native Quantum Circuit Rewriting Nov 2025

Co-author; arXiv preprint

- Encoded ZX-calculus rewrites as openCypher queries in Memgraph, achieving up to an order-of-magnitude speedup for independent rewrites over PyZX.
- Integrated tensor- and graph-equivalence checks to validate rewrite correctness and expose pattern-matching bottlenecks in graph database engines.

Selected Publications

Vista: Verifier-in-the-Loop Agentic Reinforcement Learning for Quantum Program Synthesis May 2026
Cong Yu, Tuo Shi, Valter Uotila, Shilong Deng, Lei You, Bo Zhao
10.1145/3786335.3813148 (ACM CAIS 2026, pp. 239-252)

SHARP: Shared State Reduction for Efficient Matching of Sequential Patterns May 2026
Cong Yu, Tuo Shi, Matthias Weidlich, Bo Zhao
10.14778/3796195.3796210 (Proceedings of the VLDB Endowment 19(5), pp. 987-1000)

Agent-Q: Fine-Tuning Large Language Models for Quantum Circuit Generation and Optimization Sept 2025
Linus Jern, Valter Uotila, **Cong Yu**, Bo Zhao
10.1109/QCE65121.2025.00179 (IEEE International Conference on Quantum Computing and Engineering (QCE), pp. 1621-1632)

QUASAR: Quantum Assembly Code Generation Using Tool-Augmented LLMs via Agentic RL Oct 2025
Cong Yu, Valter Uotila, Shilong Deng, Qingyuan Wu, Tuo Shi, Songlin Jiang, Lei You, Bo Zhao
arxiv.org/abs/2510.00967 (arXiv preprint arXiv:2510.00967)

ZX-DB: A Graph Database for Quantum Circuit Simplification and Rewriting via the ZX-Calculus Nov 2025
Valter Uotila, **Cong Yu**, Bo Zhao
arxiv.org/abs/2511.13033 (arXiv preprint arXiv:2511.13033)

Teaching

- Teaching Assistant, Scalable Systems and Data Management D (CS-E4780), Aalto University
- Grader, Linear Algebra for Engineering (EE 510), University of Southern California